

Exosuit Cable Stiffness Control and Co-Contraction Mechanism

cable/cable_with_exo_springs.py

Isaac-sim robotics notes

April 2026

1 Problem: Link 2 Strikes an Object

The cup manipulator’s elbow joint q_2 is driven by antagonistic **drive cables** (green/red) via an SEA motor. During normal operation the drive cables alone govern joint motion:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \begin{bmatrix} 0 \\ \tau_2^{\text{drive}} \end{bmatrix}. \quad (1)$$

Scenario. The joint is at $q_2 = 30$ (0.524 rad) when link 2 strikes an object. The impact deflects the joint to $q_2 = 45$ — a +15 (0.262 rad) disturbance. The acceleration spike reaches $|\ddot{q}_2| = 120 \text{ rad/s}^2$.

What we need: increase the joint stiffness *instantly* to resist the deflection, without changing the equilibrium position. The drive cables cannot do this alone — they are already committed to trajectory tracking.

Solution: a second set of cables — the **exosuit cables** (orange/magenta) — that provide on-demand stiffness via co-contraction. Two methods for routing these cables are compared in this document.

Key design question (Thrish, April 2026): “Are the two purple motors driving only stiffness / co-contraction, or are they driving the joint too?”

Answer: The exo motors drive *only stiffness*. The drive cables actuate the joint. Stiffness control is independent of and additive to joint rotation control: when the exo motors are off, the springs are slack and exert zero force regardless of q_2 .

2 Physical Architecture

Cable	Route	Role	Actuator
Green (drive)	DrivePulley → Idler R → BigPulley → End L	Joint torque $+\tau_2$	SEA motor
Red (drive)	DrivePulley → Idler L → BigPulley → End R	Joint torque $-\tau_2$	SEA motor
Orange (exo R)	Start R → L1 Pulley R → Elbow Pulley R $\xrightarrow{\text{spring}}$ End R	Stiffness +	Exo motor R
Magenta (exo L)	Start L → L1 Pulley L → Elbow Pulley L $\xrightarrow{\text{spring}}$ End L	Stiffness +	Exo motor L

Table 1: Four cable routes on the cup manipulator.

The key design feature is the **spring element** between the elbow pulley and the link-2 anchor. When the exo motors are off, the springs sit at their rest length, so $F_{\text{exo}} = 0$ regardless of q_2 . This provides *mechanical decoupling*: stiffness stays unchanged during free rotation.

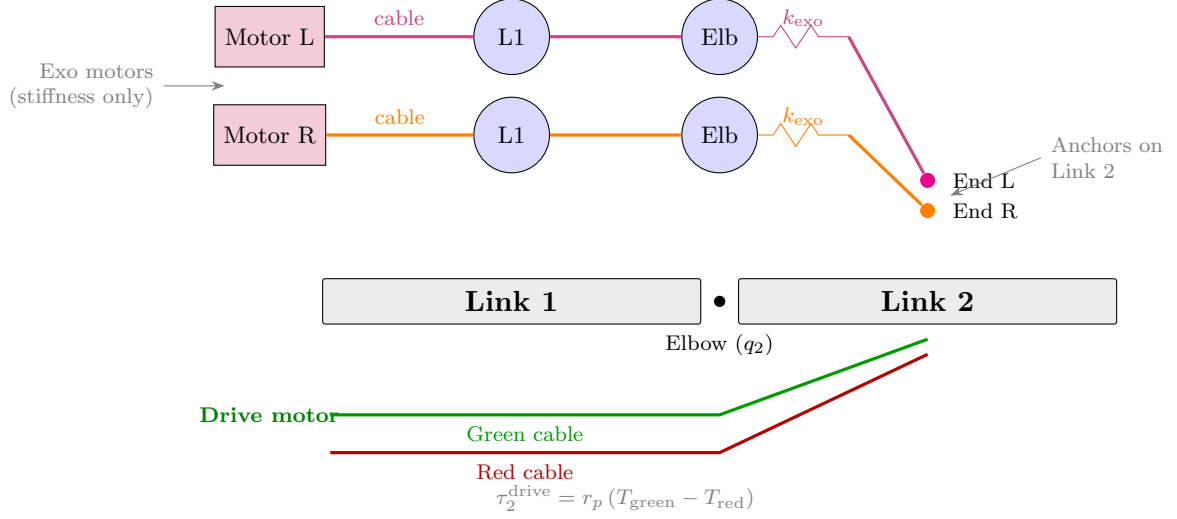


Figure 1: Dual cable architecture. Drive cables (bottom, green/red) actuate the elbow. Exo cables (top, orange/magenta) route through link-1 and elbow pulleys with a spring element before the link-2 anchor.

3 Parameters and Notation

4 Method A: Offset Pulleys (Current Design)

4.1 Cable geometry and effective moment arm

Each exo cable wraps around a **Link-1 pulley** ($r_{L1} = 41$ mm) and an **Elbow pulley** ($r_{elb} = 32$ mm, centred at $x = 232$ mm on the q_1 body). The elbow *joint axis* is at $x = 335$ mm, giving an offset:

$$d_{\text{off}} = 335 - 232 = 103 \text{ mm.} \quad (2)$$

Because the pulley is *not* at the joint axis, the exact cable length from pulley exit to link-2 anchor has two q_2 -dependent contributions:

$$l_c(q_2) = \underbrace{r_{\text{elb}} \alpha(q_2)}_{\text{cable wrap on elbow pulley}} + \underbrace{\|\mathbf{p}_{\text{exit}}(q_2) - \mathbf{p}_{\text{anchor}}(q_2)\|}_{\text{straight segment: pulley exit} \rightarrow \text{link-2 anchor}}, \quad (3)$$

where $\alpha(q_2)$ is the contact-arc angle on the pulley. The straight-segment length varies non-linearly because the link-2 anchor swings about the joint axis while the pulley centre is fixed at $d_{\text{off}} = 103$ mm away on link 1.

The **effective moment arm** of each exo cable is the derivative of the cable-length function:

$$l'_c(q_2) \triangleq \frac{dl_c}{dq_2}. \quad (4)$$

This quantity varies with q_2 (it is *not* a constant like a coaxial pulley radius). The tangent-geometry code in `cable_with_exo_springs.py` evaluates l_c and l'_c numerically for any q_2 .

First-order approximation (used in the worked example). For moderate joint angles the cable departs the elbow pulley nearly tangentially and the straight-segment length changes linearly, giving:

$$l_c(q_2) \approx r_{\text{elb}} q_2 \quad \implies \quad l'_c \approx r_{\text{elb}} = 32 \text{ mm.} \quad (5)$$

The error is negligible for $|q_2| < 45$ but grows at larger angles. For exact tracking a lookup table

Symbol	Description	Value	Source
<i>Link geometry</i>			
L_1	Link 1 length	334.8 mm	config JSON
L_2	Link 2 length	190.0 mm	config JSON
<i>Drive cable system</i>			
r_p	Drive pulley pitch radius	47.75 mm	HTD 5M 60T
<i>Method A: offset pulleys</i>			
r_{L1}	Exo L1 pulley outer radius	41.0 mm	<code>cable_with_exo_springs.py</code>
r_{elb}	Exo elbow pulley radius	32.0 mm	<code>cable_with_exo_springs.py</code>
d_{off}	Elbow pulley offset from joint	≈ 103 mm	335 – 232
d_a	Anchor distance from joint on L2	from CAD	<code>ExoEndRight.vis_xyz</code>
<i>Method B: centred pulley (Chris)</i>			
r_{cp}	Centred pulley radius ($= r_p$)	47.75 mm	1:1 ratio with drive
θ_m	Stepper motor angle	variable	encoder-tracking
$\Delta\theta$	Angular offset for stiffness	variable	control input
<i>Spring parameters (shared)</i>			
k_{exo}	Exo cable spring stiffness	200 N/m	design param
b_{exo}	Exo cable spring damping	2 N s/m	design param
Δl	Pretension pull distance	5 mm	co-contract cmd
$l_c(q_2)$	Exact cable-length function	eq. (3)	offset geometry
$l'_c(q_2)$	Effective moment arm $= dl_c/dq_2$	$\approx r_{elb}$	eq. (4)

Table 2: Notation and actual robot values from the CAD model.

from `cable_with_exo_springs.py` replaces the linear form (see section 6.1).

4.2 Exact tangent geometry

We now derive $l_c(q_2)$ and $l'_c(q_2)$ in closed form. Place the **2-D coordinate frame at the elbow joint axis** O_j , with link 1 along $-x$ (fig. 2).

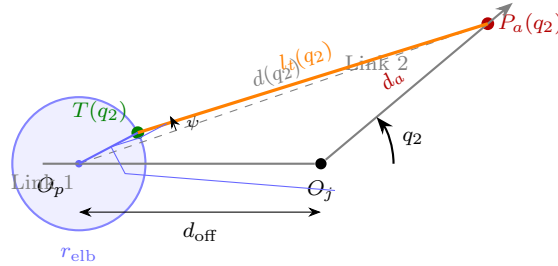


Figure 2: Tangent geometry for the offset elbow pulley. The cable exits the pulley at the tangent point $T(q_2)$ and runs straight to the anchor $P_a(q_2)$ on link 2. The straight segment is exactly the tangent from P_a to the circle (O_p, r_{elb}) . Both T and the tangent length l_t change with q_2 .

Known quantities.

- Pulley centre: $O_p = (-d_{off}, 0)$, with $d_{off} = 103$ mm.
- Pulley radius: $r_{elb} = 32$ mm.
- Anchor on link 2, at distance d_a from the joint:

$$P_a(q_2) = d_a (\cos q_2, \sin q_2).$$

Step 1 — distance from anchor to pulley centre.

$$d(q_2) = \|P_a - O_p\| = \sqrt{d_a^2 + d_{\text{off}}^2 + 2 d_a d_{\text{off}} \cos q_2}. \quad (6)$$

Step 2 — tangent contact angle. The straight cable segment is the tangent from the external point P_a to the circle (O_p, r_{elb}) . The radius to the tangent point T is perpendicular to the tangent line, giving:

$$\psi(q_2) = \arcsin\left(\frac{r_{\text{elb}}}{d(q_2)}\right). \quad (7)$$

Step 3 — tangent length (straight segment). By Pythagoras on the right triangle O_p – T – P_a :

$$l_t(q_2) = \sqrt{d(q_2)^2 - r_{\text{elb}}^2}. \quad (8)$$

Step 4 — tangent point location. Let $\varphi(q_2) = \text{atan2}(d_a \sin q_2, d_a \cos q_2 + d_{\text{off}})$ be the bearing from O_p to P_a . The tangent point on the pulley is:

$$T(q_2) = O_p + r_{\text{elb}} (\cos(\varphi + \psi), \sin(\varphi + \psi)), \quad (9)$$

where the sign of ψ selects the branch (right vs. left cable). This is exactly the construction used in `PulleyBase.compute_tangent()` in `cable/pulley.py`.

Step 5 — wrap arc. Let φ_{in} be the bearing at which the cable arrives from the Link-1 pulley (fixed, independent of q_2). The wrap angle is:

$$\alpha(q_2) = \varphi_{\text{in}} - (\varphi(q_2) + \psi(q_2)) \pmod{2\pi, \text{ taken positive}}. \quad (10)$$

Step 6 — total cable length and moment arm. Substituting into (3):

$$l_c(q_2) = r_{\text{elb}} \alpha(q_2) + \sqrt{d(q_2)^2 - r_{\text{elb}}^2} \quad (11)$$

The effective moment arm is:

$$l'_c(q_2) = \frac{dl_c}{dq_2} = r_{\text{elb}} \alpha'(q_2) + \frac{d(q_2) d'(q_2)}{\sqrt{d(q_2)^2 - r_{\text{elb}}^2}}, \quad (12)$$

where $d'(q_2) = -d_a d_{\text{off}} \sin q_2 / d(q_2)$ from (6). Both terms depend on q_2 : the wrap-rate term $\alpha'(q_2)$ and the tangent-length rate.

Why the approximation $l'_c \approx r_{\text{elb}}$ works at small angles. When $d_{\text{off}} \ll d_a$ (or equivalently $q_2 \rightarrow 0$), the distance d is nearly constant, ψ barely changes, and the tangent length changes slowly. The wrap-arc term $r_{\text{elb}} \alpha'$ dominates, and since $\alpha \approx q_2$ for small angles, $l'_c \approx r_{\text{elb}}$. At larger angles the second term in (12) grows, and l'_c deviates from the constant r_{elb} .

4.3 Spring extension and torque

With motor cable displacements $l_{m,R}$ and $l_{m,L}$, the spring extensions are (positive $q_2 = \text{CCW}$):

$$\delta_R = l_{m,R} - l_c(q_2), \quad \delta_L = l_{m,L} + l_c(q_2). \quad (13)$$

Spring force (tension only): $F_i = \max(k_{\text{exo}} \delta_i + b_{\text{exo}} \dot{\delta}_i, 0)$.

By virtual work, each cable contributes torque through the effective moment arm $l'_c(q_2)$, giving the net exo torque on the elbow:

$$\tau_2^{\text{exo},A} = l'_c(q_2) (-F_R + F_L) \quad (14)$$

4.4 Co-contraction: symmetric pull for pure stiffness

When a disturbance is detected ($|\ddot{q}_2| > 50 \text{ rad/s}^2$), the controller captures q_2^{nom} (the pre-disturbance angle) and commands both motors symmetrically with pretension Δl :

$$l_{m,R} = l_c(q_2^{\text{nom}}) + \Delta l, \quad l_{m,L} = -l_c(q_2^{\text{nom}}) + \Delta l. \quad (15)$$

Substituting into (13) and ignoring damping. Define $\Delta l_c = l_c(q_2) - l_c(q_2^{\text{nom}})$ as the cable-length change due to the disturbance:

$$F_R = k_{\text{exo}}(-\Delta l_c + \Delta l), \quad F_L = k_{\text{exo}}(+\Delta l_c + \Delta l). \quad (16)$$

Net torque via (14):

$$\tau_2^{\text{exo,A}} = l'_c(q_2)(-F_R + F_L) = \boxed{2 k_{\text{exo}} l'_c(q_2) \Delta l_c} \quad (17)$$

The pretension Δl **cancels** — it creates stiffness only, no net torque at $q_2 = q_2^{\text{nom}}$. Effective stiffness increase (linearising $\Delta l_c \approx l'_c \Delta q$ for small deflections):

$$k_{\text{eff}}^{\text{A}} = 2 k_{\text{exo}} [l'_c(q_2)]^2 \quad (18)$$

This is the *exact* stiffness formula; it varies with q_2 because l'_c is not constant for an offset pulley.

4.5 Worked example: 30° impact to 45°

For numerical evaluation we use the first-order approximation $l_c(q_2) \approx r_{\text{elb}} q_2$ and $l'_c \approx r_{\text{elb}} = 32 \text{ mm}$ (eq. (5)), which is accurate for $|q_2| < 45$.

T₀: Joint at $q_2 = 30$. Exo **OFF**. $F_R = F_L = 0$.

T₁: Impact deflects joint to 45. Acceleration spike triggers co-contraction. Controller captures $q_2^{\text{nom}} = 30$.

T₂: Motors command via (15) with $l_c(q_2^{\text{nom}}) \approx r_{\text{elb}} \times 0.524 = 0.0168 \text{ m}$:

$$l_{m,R} = 0.0168 + 0.005 = 0.0218 \text{ m}, \quad l_{m,L} = -0.0168 + 0.005 = -0.0118 \text{ m}.$$

Spring extensions at $q_2 = 45$ (0.785 rad), using $l_c(0.785) \approx 0.032 \times 0.785 = 0.0251 \text{ m}$:

$$\begin{aligned} \delta_R &= 0.0218 - 0.0251 = -0.0033 \text{ m} \Rightarrow F_R = 0 \text{ (slack)}, \\ \delta_L &= -0.0118 + 0.0251 = +0.0133 \text{ m} \Rightarrow F_L = 2.67 \text{ N}. \end{aligned}$$

$$\text{Restoring torque (moment arm } l'_c \approx r_{\text{elb}}): \tau_2^{\text{exo,A}} = 0.032 \times 2.67 = \boxed{+0.085 \text{ N m}}$$

Note: Only the L cable is active — the R cable went slack because $\Delta l = 5 \text{ mm}$ is insufficient for a +15 deflection. If $\Delta l > l'_c \cdot \Delta q \approx 8.4 \text{ mm}$, both cables stay taut and the full symmetric torque is 0.107 N m.

pproximate $k_{\text{eff}}^{\text{A}}$: From (18) with $l'_c \approx r_{\text{elb}}$:

$$k_{\text{eff}}^{\text{A}} \approx 2 \times 200 \times 0.032^2 = \boxed{0.41 \text{ N m/rad}}$$

5 Method B: Centred Pulley + Encoder Tracking (Chris)

Chris Kalogroulis (April 2026) proposed replacing the two offset elbow pulleys with **one pulley centred on the elbow joint axis** ($r_{\text{cp}} = r_p = 47.75 \text{ mm}$).

Fundamental operating principle. Because the exo pulley sits *on* the joint axis, any rotation of q_2 wraps or unwraps cable around it by exactly $r_{\text{cp}} \Delta q_2$. Without active compensation the springs would stretch, producing undesired restoring forces during normal motion.

The solution is a **stepper motor with encoder feedback** that continuously feeds (or retracts) cable at the same rate the joint moves:

- **Exo OFF** ($\Delta\theta = 0$): The servo tracks q_2 so that $\theta_m = q_2$ (right cable) or $\theta_m = -q_2$ (left cable). This keeps each spring at its rest length, producing *zero force* regardless of joint angle — the cable is neither slack nor under tension.
- **Exo ON** ($\Delta\theta > 0$): The controller adds a symmetric angular offset $\Delta\theta$ to both motor commands. Each spring is extended by $r_{\text{cp}} \Delta\theta$, creating equal-and-opposite pretension. The net torque at equilibrium is zero, but any deviation from q_2 produces a restoring torque proportional to $k_{\text{eff}}^{\text{B}}$.

Contrast with Method A: Method A uses springs on offset pulleys: when the exo motors are off the springs sit at rest *passively* (no encoder needed) because the cable geometry does not couple joint rotation to spring extension. In Method B the encoder tracking is **essential** even when no stiffness is desired — without it, joint rotation alone would load the springs.

5.1 Exact cable kinematics (no approximation)

Why Method B requires no geometric approximation. Apply the same tangent decomposition as Method A (section 4.2), but now with $d_{\text{off}} = 0$ (pulley centre $O_p \equiv$ joint axis O_j).

1. **Distance.** From (6) with $d_{\text{off}} = 0$: $d(q_2) = d_a$ — constant for all q_2 .
2. **Straight (tangent) segment.** From (8): $l_t = \sqrt{d_a^2 - r_{\text{cp}}^2}$ — also constant. The tangent point T and the anchor P_a rotate *together* around $O_p = O_j$, so their separation never changes.
3. **Wrap arc.** The bearing from O_p to P_a increases at exactly $\dot{q}_2$, so the wrap angle changes linearly: $\alpha(q_2) = \alpha_0 + q_2$.

Combining:

$$l_c^{\text{B}}(q_2) = \underbrace{r_{\text{cp}}(\alpha_0 + q_2)}_{\text{wrap arc}} + \underbrace{\sqrt{d_a^2 - r_{\text{cp}}^2}}_{\text{straight segment (constant)}} \quad (19)$$

Everything except the $r_{\text{cp}} q_2$ term is constant, so the *change* in cable length is:

$$\Delta l_c^{\text{B}} = r_{\text{cp}} \Delta q_2 \quad \implies \quad (l_c^{\text{B}})' = r_{\text{cp}} \quad (\text{exact, constant for all } q_2) \quad (20)$$

Contrast with Method A: the offset ($d_{\text{off}} \neq 0$) makes $d(q_2)$ vary, so both the straight segment *and* the wrap rate depend on q_2 . In Method B that complication vanishes because $d_{\text{off}} = 0$.

5.2 Encoder tracking and stiffness via angular offset

Each stepper motor continuously tracks the joint encoder, feeding cable at the rate the joint moves:

$$\theta_{m,R} = q_2 + \Delta\theta, \quad \theta_{m,L} = -q_2 + \Delta\theta. \quad (21)$$

The q_2 term keeps cables taut (zero extension, zero force during free rotation). The $\Delta\theta$ term creates symmetric pretension.

Deriving the spring extension (eq 22) from the general form (eq 13). The general spring extension for the right cable is $\delta_R = l_{m,R} - l_c(q_2)$. Three substitutions turn this into the angular form:

1. **Cable kinematics** (eq. (20)): $l_c(q_2) = r_{cp} q_2 + \text{const.}$
2. **Motor cable output.** The exo motor drives a pulley of the same radius r_{cp} (1:1 ratio), so $l_{m,R} = r_{cp} \theta_{m,R} + \text{const}$ (the same constant, set at assembly so that $\delta = 0$ when $\theta_m = q_2$).
3. **Substitute both into the general form:**

$$\delta_R = r_{cp} \theta_{m,R} - r_{cp} q_2 = r_{cp} (\theta_{m,R} - q_2).$$

4. **Insert encoder-tracking law** (eq. (21)), $\theta_{m,R} = q_2 + \Delta\theta$:

$$\delta_R = r_{cp} ((q_2 + \Delta\theta) - q_2) = r_{cp} \Delta\theta.$$

The q_2 terms cancel — this is the whole point of encoder tracking. Repeating for the left cable ($\theta_{m,L} = -q_2 + \Delta\theta$, $\delta_L = l_{m,L} - l_c = r_{cp}(\theta_{m,L} + q_2)$):

$$\delta_L = r_{cp} ((-q_2 + \Delta\theta) + q_2) = r_{cp} \Delta\theta.$$

Result:

$$\delta_R = r_{cp} \Delta\theta, \quad \delta_L = r_{cp} \Delta\theta \quad (\text{independent of } q_2). \quad (22)$$

Because the tracking is exact (eq. (19)), the motor compensates $r_{cp} q_2$ of cable per radian with *zero* geometric error, and any residual torque during free rotation is exactly zero.

5.3 Disturbance response

When a disturbance pushes the joint from q_2^{nom} to $q_2^{\text{nom}} + \Delta q$ before the stepper catches up:

$$F_R = k_{\text{exo}} r_{cp} (\Delta\theta - \Delta q), \quad F_L = k_{\text{exo}} r_{cp} (\Delta\theta + \Delta q). \quad (23)$$

Net torque (moment arm = r_{cp}):

$$\tau_2^{\text{exo,B}} = r_{cp} (-F_R + F_L) = \boxed{2 k_{\text{exo}} r_{cp}^2 \Delta q} \quad (24)$$

Same structure as Method A (eq. (17)), but r_{cp} replaces r_{elb} . Effective stiffness increase:

$$k_{\text{eff}}^{\text{B}} = 2 k_{\text{exo}} r_{cp}^2 = 2 \times 200 \times 0.04775^2 = \boxed{0.91 \text{ N m/rad}} \quad (25)$$

5.4 Worked example: same 30° impact to 45°

T₀: Joint at $q_2 = 30$. Stepper tracking encoder: $\theta_{m,R} = 0.524 + 0.15 = 0.674$ rad. Both springs pretensioned by $r_{cp} \cdot \Delta\theta = 7.16$ mm.

T₁: Impact deflects joint to 45. Stepper still at $\theta_{m,R} = 0.674$ rad.

T₂: Spring extensions:

$$\begin{aligned} \delta_R &= 0.04775 \times (0.674 - 0.785) = -0.00534 \text{ m} \Rightarrow F_R = 0 \text{ (slack)}, \\ \delta_L &= 0.04775 \times (-0.524 + 0.15 + 0.785) = +0.01963 \text{ m} \Rightarrow F_L = 3.93 \text{ N}. \end{aligned}$$

$$\text{Restoring torque: } \tau_2^{\text{exo,B}} = 0.04775 \times 3.93 = \boxed{+0.188 \text{ N m}}$$

2.2× more restoring torque than Method A.

T₃: Stepper catches up (~ 5 ms), re-centres around $q_2^{\text{nom}} = 30$ with offset $\Delta\theta$. Both cables taut; full symmetric stiffness: $\tau = 0.91 \times 0.262 = 0.239$ N m.

6 Comparison of Both Methods

Stiffness ratio (at moderate angles). Using the first-order approximation $l'_c \approx r_{\text{elb}}: r_{\text{cp}}/r_{\text{elb}} = 47.75/32 = 1.49\times$. Because stiffness scales as the moment arm squared:

$$\frac{k_{\text{eff}}^{\text{B}}}{k_{\text{eff}}^{\text{A}}} \approx \frac{r_{\text{cp}}^2}{r_{\text{elb}}^2} = \left(\frac{47.75}{32}\right)^2 = 2.23\times$$

Method B provides more than **twice** the effective stiffness for the same spring constant, because $r_{\text{cp}} > l'_c$ for all q_2 .

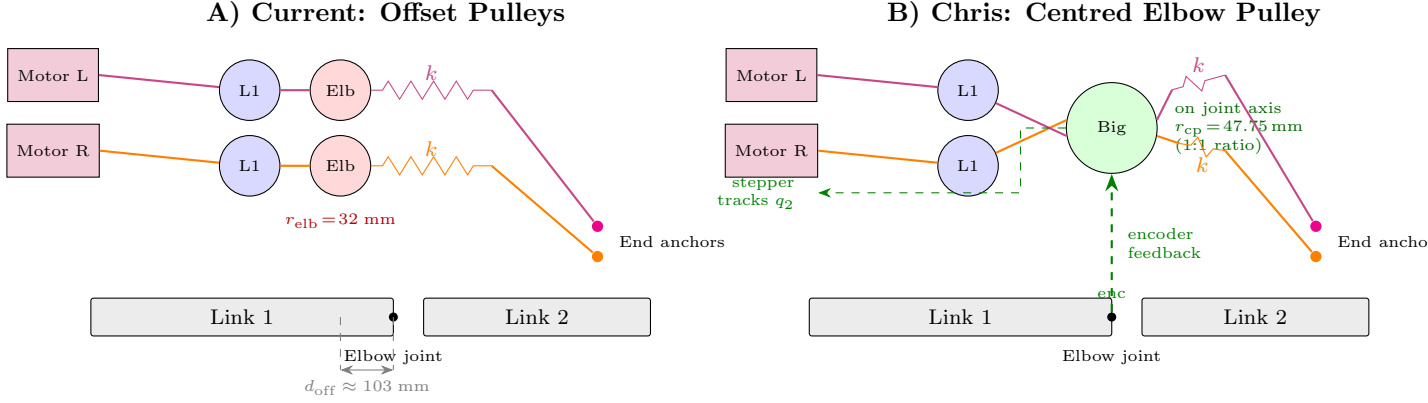


Figure 3: Routing comparison. Both methods share motor pulleys and L1 pulleys. **(A)** Two offset elbow pulleys ($r_{\text{elb}} = 32 \text{ mm}$, 103 mm before joint); cable payout $l_c(q_2)$ (exact, nonlinear; $\approx r_{\text{elb}} q_2$ at moderate angles). **(B)** One centred elbow pulley ($r_{\text{cp}} = 47.75 \text{ mm}$, on joint axis); cable payout $= r_{\text{cp}} q_2$ (exact).

Quantity	Method A (offset)	Method B (centred)
Pulley radius	$r_{\text{elb}} = 32 \text{ mm}$	$r_{\text{cp}} = 47.75 \text{ mm}$
Cable kinematics	$l_c(q_2)$ (eq. (3))	$l = r_{\text{cp}} q_2$ (exact)
Effective stiffness k_{eff}	$2 k_{\text{exo}} [l'_c]^2$ (≈ 0.41 at moderate q_2)	0.91 N m/rad
Restoring torque ($\Delta q = 15$)	$0.085\text{--}0.107 \text{ N m}$	$0.188\text{--}0.239 \text{ N m}$
Torque during free rotation	Small residual (geometry error)	Exactly zero
Cable slack management	Passive (springs absorb)	Active (stepper tracks encoder)
Slack risk	Low (springs provide compliance)	Depends on stepper latency

Table 3: Quantitative comparison. Both use $k_{\text{exo}} = 200 \text{ N/m}$.

Aspect	Method A (offset pulleys)	Method B (encoder-tracking)
Pulleys per cable	2 (L1 + offset Elbow)	2 (L1 + centred Elbow at joint)
Bandwidth requirement	Low (slow activation)	High (real-time tracking)
Single point of failure	Springs degrade gracefully	Encoder/stepper failure $\rightarrow$ slack
Prototyping effort	Springs + CubeMars	Steppers + encoder only
Large-angle accuracy	Needs lookup table	Exact ($r_{\text{cp}} \cdot q_2$)

Table 4: Design trade-offs between the two routing approaches.

6.1 Method A upgrade: adding encoder tracking (A+)

Method A can also use the encoder. The motor commands track the exact cable-length function from (3):

$$l_{m,R}(t) = l_c(q_2(t)) + \Delta l, \quad l_{m,L}(t) = -l_c(q_2(t)) + \Delta l. \quad (26)$$

This keeps both cables taut at all times (no more single-cable slack), enables always-on variable stiffness, and eliminates the activation threshold. On a real-time controller, $l_c(q_2)$ is evaluated via a lookup table from the tangent-geometry code in `cable_with_exo_springs.py`; the approximation $l_c \approx r_{\text{elb}} q_2$ (eq. (5)) may be used for moderate angles.

	A: Offset, no encoder	A+: Offset, with encoder	B: Centred, with encoder
Motor command	Fixed l_m (on trigger)	$l_c(q_2(t)) + \Delta l$	$r_{\text{cp}} \cdot q_2(t) + r_{\text{cp}} \cdot \Delta \theta$
Cables taut?	One may go slack	Both (if Δl sufficient)	Both (if $\Delta \theta$ sufficient)
Stiffness	0–0.41 N m/rad	0.41 N m/rad	0.91 N m/rad
Activation	Threshold	Always on	Always on
Geometric accuracy	N/A	$l_c(q_2)$ lookup table	Exact

Table 5: Three operating modes. “A+” combines springs with encoder feedback.

Bottom line. In Method B the encoder is **essential** (cables go slack without it). In Method A it is an **upgrade** that eliminates single-cable slack and enables always-on variable stiffness. The trade-off: mechanical simplicity (Method B, exact $r \cdot q_2$) versus robustness (Method A+, springs catch errors, but need a lookup table for large angles).

Prototype strategy (Chris): Build a quick first exosuit with *only* stepper motors and encoder — no springs, no CubeMars. If steppers track the encoder with < 5 ms lag, the full system is viable.

7 Dynamics and Control Summary

Full elbow dynamics with both cable systems:

$$M_{22} \ddot{q}_2 + c_2(\mathbf{q}, \dot{\mathbf{q}}) + g_2(\mathbf{q}) = \underbrace{r_p (T_{\text{green}} - T_{\text{red}})}_{\text{drive cables}} + \underbrace{r_{\text{eff}}(q_2) (-F_R + F_L)}_{\text{exo cables (when active)}} \quad (27)$$

where the effective moment arm is $r_{\text{eff}} = l'_c(q_2)$ (Method A, varies with q_2) or $r_{\text{eff}} = r_{\text{cp}}$ (Method B, constant).

Aspect	Biological	Our mechanism
Agonist/Antagonist	Biceps/Triceps	Drive green/red cables
Co-contraction	Both muscles activate	Both exo motors pull
Stiffness source	Muscle short-range stiffness	Exo spring k_{exo}
Activation trigger	Reflex (stretch reflex)	$ \ddot{q}_2 > \text{threshold}$
Decoupling	Reciprocal inhibition	Spring at rest $\Rightarrow F = 0$

Table 6: Biological co-contraction analogy.

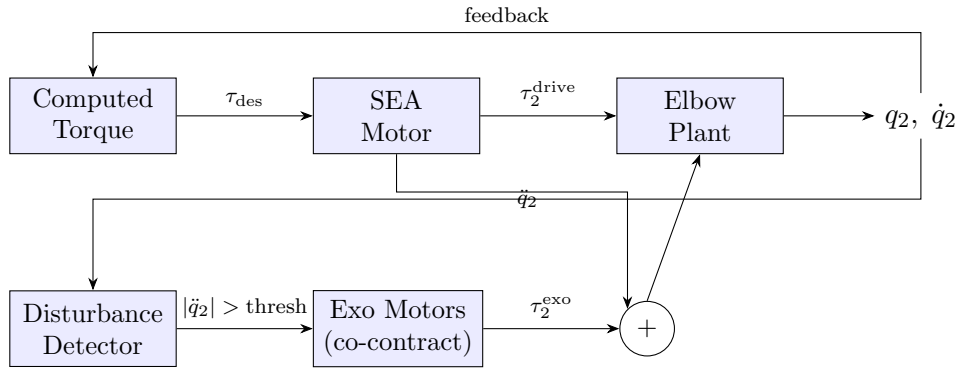


Figure 4: Control block diagram. Drive path (top) runs continuously; exo path (bottom) activates on disturbance detection.